PHYSICAL CHEMISTRY CUMULATIVE EXAM

January 21, 1994

Is the orbital motion of the earth around the sun a case of perpetual motion and hence a violation of the law of thermodynamics? Explain.

Is the ceaseless, chaotic motion of molecules in a gas a case of perpetual motion and hence a violation of the laws of thermodynamics? Explain.

A person who has a superficial acquaintance with thermodynamics consults you about the following difficulty: "For the reaction $2H_2 + O_2 \rightarrow 2H_2O$, ΔS is negative as we can see in the tables of third-law entropies. Entropy has to increase in every process. Should we conclude that this reaction is impossible?" Explain what the incorrect assumption is, what the corresponding correct statement is, and the special conditions, if any, under which the original assumption and conclusion would become correct.

A person who has a superficial acquaintance with thermodynamics consults you about the following difficulty: "I thought that every spontaneous process is accompanied by an increase of entropy, but on calculating the entropy change for the freezing of 1 mole of supercooled water at -5°C, which I know intuitively to be a spontaneous process, I obtain an entropy change of $-5.04 cal°K^{-1} mole^{-1}$ for the H_2O. Since this is a spontaneous process, why isn't the entropy change positive?" Explain what the incorrect assumption is, what the corresponding correct statement is, and the special conditions, if any, under which the original assumption and conclusion would become correct.